

PRIM'S ALGORITHM

```

# Prim's Algorithm

INF = 9999999

# Number of vertices
V = 5

# Graph represented using adjacency matrix
graph = [
    [0, 2, 0, 6, 0],
    [2, 0, 3, 8, 5],
    [0, 3, 0, 0, 7],
    [6, 8, 0, 0, 9],
    [0, 5, 7, 9, 0]
]

selected = [False] * V

# Start from vertex 0
selected[0] = True

print("Edges in Minimum Spanning Tree:")

edges = 0
total_cost = 0

while edges < V - 1:
    minimum = INF
    x = 0
    y = 0

    for i in range(V):
        if selected[i]:
            for j in range(V):
                if not selected[j] and graph[i][j] != 0:
                    if graph[i][j] < minimum:
                        minimum = graph[i][j]
                        x = i
                        y = j

    print(chr(65 + x), "-", chr(65 + y), ":", minimum)

    total_cost += minimum
    selected[y] = True
    edges += 1

print("Total cost of MST:", total_cost)

```

```

Edges in Minimum Spanning Tree:
A - B : 2
B - C : 3
B - E : 5
A - D : 6
Total cost of MST: 16

```